

CANBERRA TUGGERANONG, AustraliaWMO: **949250**Lat: **35.4183S**Lon: **149.0936E**Elev: **1927**StdP: **13.70**Time Zone: **10.00 (AUS)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	27.3	22.4	18.1	34.5	24.4	19.9	35.9	20.1	50.3	18.1	48.7	0.4	100	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.3	93.7	65.3	89.7	64.5	86.1	63.2	69.0	83.2	67.4	80.6	66.0	78.9	11.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	102.1	70.6	63.7	94.9	70.4	61.9	88.8	70.1	34.3	83.4	32.9	80.1	31.8	78.8	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	16.0	14.3	DB	21.0	99.6	1.4	3.3	19.9	102.0	19.1	103.9	18.3	105.7	17.2	108.1	(4)
(5)			WB	20.2	71.1	1.4	1.8	19.2	72.4	18.3	73.4	17.5	74.4	16.5	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.5	71.1	69.1	64.5	56.8	48.9	44.6	43.0	45.0	50.6	56.1	61.9	66.8	(6)	
	DBStd	10.97	6.08	5.60	5.35	5.11	4.67	4.33	4.25	4.08	4.73	5.56	6.27	5.72	(7)	
	HDD50	698	0	0	0	7	78	168	222	164	49	9	1	0	(8)	
	HDD65	3667	17	20	75	251	499	611	683	620	433	279	132	47	(9)	
	CDD50	3056	653	536	449	210	44	7	4	9	66	199	358	521	(10)	
	CDD65	550	205	136	59	4	0	0	0	0	4	39	103		(11)	
	CDH74	6452	2283	1335	630	76	0	0	0	0	6	166	649	1307	(12)	
	CDH80	2554	1093	533	176	10	0	0	0	0	0	30	215	497	(13)	
(14) Wind	WSAvg	5.7	6.3	5.9	5.7	4.6	4.5	4.7	5.2	5.9	6.2	6.4	6.3	6.5	(14)	
(15) Precipitation	PrecAvg	27.6	2.7	2.8	1.9	1.6	1.5	2.6	2.1	2.3	2.6	2.4	2.8	2.6	(15)	
	PrecMax	43.9	8.5	7.8	6.8	6.8	6.0	7.0	5.2	5.4	6.2	5.4	6.7	7.0	(16)	
	PrecMin	16.1	0.1	0.4	0.2	0.1	0.2	0.6	0.7	0.4	0.2	0.4	0.4	0.5	(17)	
	PrecStd	6.1	1.8	2.0	1.4	1.5	1.3	1.5	1.1	1.2	1.3	1.2	1.7	1.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.7	97.2	89.8	81.4	69.4	60.8	59.8	65.5	74.3	84.2	91.6	96.0	(19)	
		MCWB	66.6	68.0	63.2	59.1	55.0	50.7	50.3	51.4	55.6	58.8	63.1	64.0	(20)	
	2%	DB	95.3	90.7	84.8	75.6	65.8	57.9	56.8	61.1	70.1	78.7	86.0	90.0	(21)	
		MCWB	66.1	65.2	62.4	58.2	53.8	49.3	47.8	49.4	54.5	57.7	60.9	63.4	(22)	
	5%	DB	91.0	86.5	80.8	72.1	62.8	55.9	54.6	58.2	66.6	74.5	81.4	85.7	(23)	
		MCWB	65.6	64.5	61.6	57.2	52.3	48.6	46.6	48.1	52.7	56.4	60.2	62.3	(24)	
	10%	DB	86.4	82.3	77.0	68.7	60.0	53.9	52.6	55.6	63.3	70.1	76.8	81.5	(25)	
		MCWB	64.5	63.2	60.7	56.0	51.3	47.7	45.5	46.6	51.3	55.2	59.0	61.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	71.6	68.3	63.0	59.7	54.1	53.6	54.4	58.3	62.5	66.6	68.6	(27)	
		MCDB	88.2	84.0	76.8	73.2	63.2	55.9	56.4	60.6	68.0	73.5	79.5	79.8	(28)	
	2%	WB	69.2	69.2	66.2	60.7	55.7	51.7	50.1	51.4	56.5	60.4	64.4	66.3	(29)	
		MCDB	85.8	81.6	75.6	70.0	62.8	55.4	54.0	57.9	65.6	71.0	78.2	80.3	(30)	
	5%	WB	67.7	67.2	64.5	58.9	53.8	49.9	48.0	49.4	54.7	58.5	62.6	64.8	(31)	
		MCDB	83.4	78.8	74.1	67.5	60.6	54.2	52.5	56.4	63.5	69.3	74.6	78.6	(32)	
	10%	WB	66.3	65.5	62.7	57.4	52.0	48.4	46.3	47.5	52.6	56.7	60.9	63.3	(33)	
		MCDB	80.2	77.7	73.2	66.3	58.8	53.0	51.6	54.5	61.6	67.4	72.7	76.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	25.3	22.6	22.5	23.5	23.3	19.3	20.0	21.8	23.5	24.1	24.2	24.6	(35)	
		MCDBR	33.7	30.8	29.8	29.4	27.9	22.7	23.3	26.7	30.5	32.8	33.1	33.1	(36)	
		MCWBR	9.9	9.8	11.1	13.8	16.4	14.8	15.2	16.2	16.4	14.6	12.3	10.7	(37)	
	5% WB	MCDBR	28.0	23.6	21.7	23.3	23.5	18.1	18.1	22.8	25.7	25.3	26.0	26.0	(38)	
		MCWBR	9.6	8.9	9.7	11.8	14.6	12.5	12.7	15.0	14.6	12.6	11.0	9.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.391	0.346	0.317	0.299	0.290	0.273	0.268	0.280	0.308	0.328	0.340	0.342	(40)	
	taud		2.356	2.530	2.592	2.614	2.580	2.648	2.660	2.604	2.519	2.482	2.494	2.501	(41)	
	Ebn at Noon		300	307	305	293	278	276	285	295	303	309	314	316	(42)	
	Edh at Noon		42	34	29	25	23	20	21	25	31	35	36	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2257	1933	1645	1257	937	717	809	1090	1511	1882	2079	2305	(44)	
(45)	RadStd		169	171	138	101	47	59	54	74	127	160	170	149	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+0.97	N/A	N/A	N/A	+1.60	+1.71	N/A	N/A	N/A	N/A
Regional (26 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+194	N/A

Nomenclature: See separate page